Statistics/Data analysis

```
      name:  The_value_of_a_helping_hand_sup
       log:  C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_3.smcl
  log type:  smcl
 opened on:   5 Apr 2026, 13:42:29
C:\Users\hansg\Documents\working\SHARE data\weighting
(1 observation deleted)
(10,402 real changes made)
(6,150 real changes made)
(70,195 real changes made)
(8,277 real changes made)
(23,812 real changes made)
(1,763 real changes made)
(1,198 real changes made)
(1,763 real changes made)

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
         age |    109,546         0.0000         0.0000             .          .

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
       thinc |    109,546         0.0000         0.0000             .          .

Skewness and kurtosis tests for normality
                                                       ------ Joint test ------
    Variable |        Obs   Pr(skewness)   Pr(kurtosis)   Adj chi2(2)  Prob>chi2
-------------+-------------------------------------------------------------------
        yedu |    109,546         0.0000         0.0000        100.33     0.0000
(11,207 real changes made)

Summary for variables: id
Group variable: htype (Household type)

           htype |         N
-----------------+----------
             1 S |     37717
          1 CNRP |      8553
           1 C2R |     62737
    1 CNRP + 1 S |        17
         Multi S |       372
    1 C2R  + 1 S |       117
  1 C2R + 1 CNRP |         2
       Multi C2R |        29
 Multi S + 1 C2R |         2
-----------------+----------
           Total |    109546
-----------------------------------------------------------------------------

  Household
       type |      Freq.     Percent        Cum.
------------+-----------------------------------
          1 |     37,717       34.43       34.43
          2 |      8,553        7.81       42.24
          3 |     62,737       57.27       99.51
          4 |         17        0.02       99.52
          5 |        372        0.34       99.86
          6 |        117        0.11       99.97
          7 |          2        0.00       99.97
          8 |         29        0.03      100.00
         10 |          2        0.00      100.00
------------+-----------------------------------
      Total |    109,546      100.00
(37,717 real changes made)
(62,737 real changes made)
```

```
note: 2.naly omitted because of collinearity.
note: 2.rhfo#0.ghto identifies no observations in the sample.
note: 2.rhfo#1.ghto identifies no observations in the sample.
note: 2.rhfo#2.ghto omitted because of collinearity.
Iteration 0:  Log likelihood = -155920.97
Iteration 1:  Log likelihood =  -129392.9
Iteration 2:  Log likelihood = -127789.68
Iteration 3:  Log likelihood = -127779.87
Iteration 4:  Log likelihood = -127779.86

Ordered logistic regression                         Number of obs =  109,546
                                                    LR chi2(37)   = 56282.21
                                                    Prob > chi2   =   0.0000
Log likelihood = -127779.86                         Pseudo R2     =   0.1805
```

sphus	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]	
age	.0863752	.0136721	6.32	0.000	.0595783	.1131721
c.age#c.age	-.0003966	.0000905	-4.38	0.000	-.000574	-.0002192
yedu	-.0051815	.0049376	-1.05	0.294	-.014859	.004496
c.yedu#c.yedu	-.0015303	.0002161	-7.08	0.000	-.0019538	-.0011068
thinc	-.1963264	.0139047	-14.12	0.000	-.223579	-.1690737
c.thinc#c.thinc	.0287186	.0061741	4.65	0.000	.0166176	.0408196
gender						
Female	-.0246782	.0126343	-1.95	0.051	-.049441	.0000846
1.covid	-.1171794	.0196753	-5.96	0.000	-.1557422	-.0786166
household						
Couple, both responding	.1730804	.0152981	11.31	0.000	.1430967	.2030641
Multiple, at least 1 responding	.2019084	.0242979	8.31	0.000	.1542853	.2495315
cjs						
Not applicable	.5647311	.0586559	9.63	0.000	.4497676	.6796946
Retired	.3279888	.0295107	11.11	0.000	.2701488	.3858287
Unemployed	.2423596	.0877806	2.76	0.006	.0703128	.4144064
Permanently sick or disabled	1.466208	.0628076	23.34	0.000	1.343108	1.589309
Homemaker	.3872612	.0361954	10.70	0.000	.3163195	.4582029
Other	.236939	.0625004	3.79	0.000	.1144404	.3594375
gali						
Limited	1.993667	.0139058	143.37	0.000	1.966412	2.020922
lifesat	-.3221656	.0190294	-16.93	0.000	-.3594624	-.2848687
c.lifesat#c.lifesat	.0005231	.0013241	0.40	0.693	-.0020722	

```
>     .0031183
                                  |
                            naly  |
                     At least 1   |  -.3699398   .0190393   -19.43   0.000    -.4072561
>    -.3326235
                  Not applicable  |          0  (omitted)
                                  |
                            rhfo  |
                     At least 1   |   .4017629   .0184634    21.76   0.000     .3655753
>     .4379505
                  Not applicable  |  -.1816697    .032546    -5.58   0.000    -.2454587
>    -.1178806
                                  |
                            ghto  |
                     At least 1   |  -.2365278   .0181813   -13.01   0.000    -.2721624
>    -.2008932
                  Not applicable  |   .0253438   .0197795     1.28   0.200    -.0134233
>     .0641108
                                  |
                       rhfo#ghto  |
          At least 1#At least 1   |  -.1925051    .033478    -5.75   0.000    -.2581207
>    -.1268894
      At least 1#Not applicable   |  -.0256236    .046185    -0.55   0.579    -.1161445
>     .0648974
            Not applicable#None   |          0  (empty)
      Not applicable#At least 1   |          0  (empty)
  Not applicable#Not applicable   |          0  (omitted)
                                  |
                         country  |
                         Austria  |    .188177   .0296532     6.35   0.000     .1300579
>     .2462962
                         Germany  |   .6576791   .0365215    18.01   0.000     .5860982
>       .72926
                     Netherlands  |   .0773056   .0360012     2.15   0.032     .0067445
>     .1478666
                           Spain  |   .7768267   .0337809    23.00   0.000     .7106173
>     .8430361
                           Italy  |   .5808434   .0327538    17.73   0.000     .5166471
>     .6450396
                          France  |   .4939397    .029585    16.70   0.000     .4359542
>     .5519252
                         Denmark  |  -.2643423   .0348748    -7.58   0.000    -.3326957
>    -.1959889
                     Switzerland  |   .0625047   .0320327     1.95   0.051    -.0002783
>     .1252876
                         Belgium  |   .1119973   .0296268     3.78   0.000     .0539299
>     .1700647
                  Czech Republic  |   .4176848   .0319373    13.08   0.000     .3550889
>     .4802807
                        Slovenia  |    .581898    .034947    16.65   0.000     .5134032
>     .6503928
------------------------------------+----------------------------------------------------------------
  ------------
                           /cut1  |  -.6876268   .5125739                      -1.692253
>     .3169995
                           /cut2  |   1.091282   .5126655                       .0864761
>     2.096088
                           /cut3  |   3.578677   .5129142                       2.573384
>     4.583971
                           /cut4  |   5.872603   .5128953                       4.867347
>      6.87786
------------------------------------+----------------------------------------------------------------
  ------------

Tests of proportional odds assumption

Number of obs            =  109,546
Number of predictors     =      37
Number of outcome levels =       5
Degrees of freedom       =     111
```

Test	chi2	P>chi2
Brant	**4104.85**	**0.000**
Likelihood-ratio	**4041.39**	**0.000**
Score	**4144.08**	**0.000**
Wald	**4168.57**	**0.000**
Wolfe-Gould	**4080.42**	**0.000**

(file **TestAssum_3.doc** not found)
Click to Open File: TestAssum_3.doc
note: **2.naly** omitted because of collinearity.
note: **2.rhfo#0.ghto** identifies no observations in the sample.
note: **2.rhfo#1.ghto** identifies no observations in the sample.
note: **2.rhfo#2.ghto** omitted because of collinearity.

Generalized Ordered Logit Estimates Number of obs = **109,528**
Wald chi2(1) = .
Prob > chi2 = .
Log pseudolikelihood = **-122345.42** Pseudo R2 = **0.1985**

(Std. err. adjusted for **6** cluste
> rs in **Qyear**)

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% con
Excellent					
age	**.1110185**	**.0793531**	**1.40**	**0.162**	**-.0445106**
>	**.2665477**				
c.age#c.age	**-.0005385**	**.0005622**	**-0.96**	**0.338**	**-.0016405**
>	**.0005634**				
yedu	**.009336**	**.0229418**	**0.41**	**0.684**	**-.0356291**
>	**.0543012**				
c.yedu#c.yedu	**-.0011983**	**.0008803**	**-1.36**	**0.173**	**-.0029238**
>	**.0005271**				
thinc	**-.0839969**	**.0298152**	**-2.82**	**0.005**	**-.1424336**
>	**-.0255603**				
c.thinc#c.thinc	**-.0170671**	**.0158671**	**-1.08**	**0.282**	**-.0481661**
>	**.014032**				
gender					
Female	**.1369228**	**.0559608**	**2.45**	**0.014**	**.0272416**
>	**.2466039**				
covid					
1	**-.0038514**	**.0123958**	**-0.31**	**0.756**	**-.0281468**
>	**.020444**				
household					
Couple, both responding	**.1551275**	**.0491943**	**3.15**	**0.002**	**.0587085**
>	**.2515465**				
Multiple, at least 1 responding	**.1678448**	**.0668102**	**2.51**	**0.012**	**.0368991**
>	**.2987905**				
cjs					
Not applicable	**.321662**	**.5422782**	**0.59**	**0.553**	**-.7411838**
>	**1.384508**				
Retired	**.2666711**	**.1361589**	**1.96**	**0.050**	**-.0001954**
>	**.5335376**				
Unemployed	**.0982249**	**.2758631**	**0.36**	**0.722**	**-.4424568**
>	**.6389066**				
Permanently sick or disabled	**.5866364**	**.1051922**	**5.58**	**0.000**	**.3804634**
>	**.7928094**				
Homemaker	**.3648983**	**.2075903**	**1.76**	**0.079**	**-.0419712**
>	**.7717677**				

```
                        Other |  -.2767604    .322236    -0.86   0.390    -.9083313
>     .3548106
                              |
                         gali |
                      Limited |   2.029552     .04316    47.02   0.000      1.94496
>     2.114145
                              |
                      lifesat |    .486523   .0849614     5.73   0.000     .3200016
>     .6530443
                              |
            c.lifesat#c.lifesat |  -.0609273    .005423   -11.23   0.000    -.0715561
>    -.0502984
                              |
                         naly |
                    At least 1 |  -.1667244    .014015   -11.90   0.000    -.1941932
>    -.1392556
                              |
                         rhfo |
                    At least 1 |   .2022918   .0581208     3.48   0.001     .0883772
>     .3162064
                Not applicable |  -.1736055   .0273492    -6.35   0.000    -.2272089
>    -.1200022
                              |
                         ghto |
                    At least 1 |   -.197295   .0932571    -2.12   0.034    -.3800756
>    -.0145144
                Not applicable |  -.0542819   .0293667    -1.85   0.065    -.1118395
>     .0032757
                              |
                     rhfo#ghto |
        At least 1#At least 1 |   -.197565   .0898952    -2.20   0.028    -.3737563
>    -.0213737
     At least 1#Not applicable |  -.1947864   .0673556    -2.89   0.004     -.326801
>    -.0627718
                              |
                      country |
                      Austria |    .716422   .0867202     8.26   0.000     .5464536
>     .8863904
                      Germany |   1.374634   .1172268    11.73   0.000     1.144874
>     1.604395
                  Netherlands |  -.1232244   .0962593    -1.28   0.200    -.3118892
>     .0654404
                        Spain |   1.517244   .0592437    25.61   0.000     1.401129
>      1.63336
                        Italy |   .6486197   .0785892     8.25   0.000     .4945876
>     .8026517
                       France |   .6672427   .0421099    15.85   0.000     .5847088
>     .7497765
                      Denmark |   .0440525   .0636142     0.69   0.489    -.0806291
>     .1687341
                  Switzerland |   .8338404   .1031776     8.08   0.000     .6316161
>     1.036065
                      Belgium |   .6721312   .0567946    11.83   0.000     .5608158
>     .7834466
                Czech Republic |   1.433087   .1736134     8.25   0.000     1.092811
>     1.773363
                     Slovenia |   1.059123   .1896714     5.58   0.000     .6873739
>     1.430872
                              |
                        _cons |  -3.418859    3.16009    -1.08   0.279    -9.612521
>     2.774803
------------------------------+----------------------------------------------------
  ____________
Very_good
                          age |   .0462407   .0312924     1.48   0.139    -.0150912
>     .1075726
                              |
                   c.age#c.age |  -.0000814    .000203    -0.40   0.688    -.0004792
>     .0003164
                              |
                         yedu |   .0026849     .01457     0.18   0.854    -.0258718
>     .0312415
```

```
                             |
                 c.yedu#c.yedu |  -.0022478    .000571    -3.94   0.000     -.003367
>    -.0011286
                             |
                         thinc |  -.1720045   .0526148    -3.27   0.001    -.2751276
>    -.0688815
                             |
             c.thinc#c.thinc |  -.0020088   .0198864    -0.10   0.920    -.0409855
>     .0369678
                             |
                       gender |
                       Female |   .0660171   .0233965     2.82   0.005     .0201607
>     .1118734
                             |
                        covid |
                            1 |  -.0559266   .0281408    -1.99   0.047    -.1110815
>    -.0007717
                             |
                    household |
      Couple, both responding |   .2227671   .0305217     7.30   0.000     .1629457
>     .2825886
Multiple, at least 1 responding |   .2634241   .0324185     8.13   0.000      .199885
>     .3269633
                             |
                          cjs |
             Not applicable |   .2274824   .3967683     0.57   0.566    -.5501692
>     1.005134
                     Retired |   .3333797   .0643671     5.18   0.000     .2072225
>     .4595369
                  Unemployed |   .2187696   .1240382     1.76   0.078    -.0243409
>       .46188
  Permanently sick or disabled |   .6590085   .1469766     4.48   0.000     .3709397
>     .9470774
                   Homemaker |   .2978985   .0522229     5.70   0.000     .1955436
>     .4002534
                       Other |  -.1728116   .3433078    -0.50   0.615    -.8456824
>     .5000592
                             |
                        gali |
                     Limited |   1.833062   .0292884    62.59   0.000     1.775658
>     1.890467
                             |
                      lifesat |   .0485612   .0929692     0.52   0.601     -.133655
>     .2307774
                             |
          c.lifesat#c.lifesat |  -.0267894   .0060425    -4.43   0.000    -.0386324
>    -.0149464
                             |
                        naly |
                  At least 1 |   .0203553   .0442683     0.46   0.646    -.0664089
>     .1071195
                             |
                        rhfo |
                  At least 1 |   .1778826   .0453393     3.92   0.000     .0890193
>     .2667459
             Not applicable |  -.1407492   .0573377    -2.45   0.014    -.2531289
>    -.0283695
                             |
                        ghto |
                  At least 1 |  -.1386541   .0302762    -4.58   0.000    -.1979943
>    -.0793139
             Not applicable |  -.0103888   .0382357    -0.27   0.786    -.0853293
>     .0645518
                             |
                   rhfo#ghto |
      At least 1#At least 1 |  -.1064383   .1464913    -0.73   0.467    -.3935561
>     .1806795
     At least 1#Not applicable |  -.1045983   .0594703    -1.76   0.079    -.2211579
>     .0119614
                             |
                     country |
                     Austria |   .2469489   .0449087     5.50   0.000     .1589295
```

```
>     .3349683
                    Germany |   1.108254   .0679277    16.32   0.000     .9751185
>      1.24139
                Netherlands |   .3867435   .0508462     7.61   0.000     .2870868
>     .4864002
                      Spain |   1.055234   .0479818    21.99   0.000     .9611918
>     1.149277
                      Italy |   .9056285   .0513735    17.63   0.000     .8049383
>     1.006319
                     France |   .8623085    .033965    25.39   0.000     .7957384
>     .9288786
                    Denmark |  -.3682184   .0608671    -6.05   0.000    -.4875158
>     -.248921
                Switzerland |   .3804336   .0384191     9.90   0.000     .3051336
>     .4557336
                    Belgium |   .5028489   .0535096     9.40   0.000     .3979719
>     .6077258
            Czech Republic  |   .9535402   .0653933    14.58   0.000     .8253717
>     1.081709
                   Slovenia |   1.004634   .1147375     8.76   0.000     .7797529
>     1.229516
                            |
                      _cons |  -1.602917   1.169872    -1.37   0.171    -3.895824
>     .6899908
------------------------------+----------------------------------------------------------------
  ___________
Good                        |
                        age |   .1444588   .0278629     5.18   0.000     .0898485
>      .199069
                            |
                 c.age#c.age |  -.0008112    .000185    -4.39   0.000    -.0011737
>    -.0004486
                            |
                       yedu |  -.0284161   .0066302    -4.29   0.000     -.041411
>    -.0154213
                            |
              c.yedu#c.yedu |  -.0006023   .0003544    -1.70   0.089    -.0012968
>     .0000922
                            |
                      thinc |  -.2298372   .0426037    -5.39   0.000     -.313339
>    -.1463354
                            |
            c.thinc#c.thinc |   .0389674   .0196749     1.98   0.048     .0004053
>     .0775296
                            |
                     gender |
                     Female |  -.0145861   .0207778    -0.70   0.483    -.0553099
>     .0261377
                            |
                      covid |
                          1 |  -.1065718   .0230132    -4.63   0.000    -.1516768
>    -.0614669
                            |
                  household |
        Couple, both responding |   .1484918   .0475602     3.12   0.002     .0552756
>      .241708
Multiple, at least 1 responding |   .2743125    .037837     7.25   0.000     .2001533
>     .3484718
                            |
                        cjs |
            Not applicable  |   .4105642   .2533232     1.62   0.105    -.0859401
>     .9070685
                    Retired |    .263732   .1045181     2.52   0.012     .0588803
>     .4685837
                 Unemployed |   .1243392   .0972431     1.28   0.201    -.0662538
>     .3149321
   Permanently sick or disabled |   1.021168   .0652107    15.66   0.000     .8933577
>     1.148979
                  Homemaker |   .2860253   .1815558     1.58   0.115    -.0698174
>     .6418681
                      Other |   .0036122   .2510631     0.01   0.989    -.4884625
>     .4956868
```

```
                                 |
                            gali |
                         Limited |   2.112057   .0167981   125.73   0.000     2.079133
>     2.144981
                                 |
                         lifesat |  -.2904049   .0613835    -4.73   0.000    -.4107143
>    -.1700955
                                 |
             c.lifesat#c.lifesat |  -.0007219   .0042355    -0.17   0.865    -.0090232
>     .0075795
                                 |
                            naly |
                       At least 1 |  -.2107887    .053395    -3.95   0.000     -.315441
>    -.1061364
                                 |
                            rhfo |
                       At least 1 |   .4024776   .0633601     6.35   0.000     .2782941
>     .5266611
                   Not applicable |  -.1570168    .053602    -2.93   0.003    -.2620747
>    -.0519588
                                 |
                            ghto |
                       At least 1 |  -.2385443   .0499079    -4.78   0.000    -.3363619
>    -.1407267
                   Not applicable |   -.004505   .0269361    -0.17   0.867    -.0572989
>     .0482888
                                 |
                        rhfo#ghto |
          At least 1#At least 1 |  -.1386985    .143438    -0.97   0.334    -.4198319
>     .1424348
      At least 1#Not applicable |  -.0048391   .0684046    -0.07   0.944    -.1389097
>     .1292315
                                 |
                         country |
                         Austria |  -.0335374   .0994305    -0.34   0.736    -.2284177
>     .1613429
                         Germany |   .2981541   .0732533     4.07   0.000     .1545803
>     .4417279
                     Netherlands |  -.1133228   .0414064    -2.74   0.006    -.1944779
>    -.0321677
                           Spain |   .5753629   .0689949     8.34   0.000     .4401353
>     .7105904
                           Italy |   .5317732   .1117682     4.76   0.000     .3127116
>     .7508348
                          France |   .1156296   .0512571     2.26   0.024     .0151676
>     .2160916
                         Denmark |  -.0751292   .0946912    -0.79   0.428    -.2607206
>     .1104622
                     Switzerland |  -.3532466   .0839733    -4.21   0.000    -.5178313
>     -.188662
                         Belgium |  -.2724308   .0344052    -7.92   0.000    -.3398638
>    -.2049977
                   Czech Republic |  -.1623325   .1261383    -1.29   0.198    -.4095591
>      .084894
                        Slovenia |   .1726334   .0910403     1.90   0.058    -.0058024
>     .3510692
                                 |
                           _cons |  -5.513567   .9813438    -5.62   0.000    -7.436966
>    -3.590169
-------------------------------+----------------------------------------------------------------
  ____________
Fair                             |
                             age |    .171959   .0446059     3.86   0.000     .0845329
>      .259385
                                 |
                     c.age#c.age |  -.0010135   .0002946    -3.44   0.001    -.0015909
>    -.0004362
                                 |
                            yedu |  -.0161641   .0089519    -1.81   0.071    -.0337095
>     .0013814
                                 |
                     c.yedu#c.yedu |  -.0010688   .0006333    -1.69   0.091    -.0023099
```

```
>     .0001724
                                  |
                           thinc  |  -.1630324   .0697911    -2.34   0.019    -.2998204
>    -.0262443
                                  |
                c.thinc#c.thinc   |   .0464695   .0424214     1.10   0.273    -.0366749
>     .1296139
                                  |
                          gender  |
                          Female  |  -.0807255   .0523758    -1.54   0.123    -.1833803
>     .0219292
                                  |
                           covid  |
                               1  |  -.1020354   .0450317    -2.27   0.023    -.1902959
>     -.013775
                                  |
                       household  |
        Couple, both responding   |   .1148802   .0484257     2.37   0.018     .0199675
>     .2097929
Multiple, at least 1 responding   |   .0838158   .0444522     1.89   0.059    -.0033089
>     .1709406
                                  |
                             cjs  |
                Not applicable    |   .4625865   .1672589     2.77   0.006     .1347651
>     .7904079
                        Retired   |   .2976405   .0908439     3.28   0.001     .1195897
>     .4756914
                      Unemployed  |  -.7229584   .1495907    -4.83   0.000    -1.016151
>    -.4297661
   Permanently sick or disabled   |   1.453473    .093266    15.58   0.000     1.270675
>     1.636271
                       Homemaker  |   .2473714   .0963906     2.57   0.010     .0584493
>     .4362935
                           Other  |   .1188542   .1213186     0.98   0.327    -.1189259
>     .3566342
                                  |
                            gali  |
                         Limited  |   2.984527   .0713012    41.86   0.000     2.844779
>     3.124275
                                  |
                          lifesat |  -.5792448   .0563587   -10.28   0.000    -.6897058
>    -.4687838
                                  |
             c.lifesat#c.lifesat  |   .0217003   .0043234     5.02   0.000     .0132265
>      .030174
                                  |
                             naly |
                       At least 1 |  -.5431529   .0847908    -6.41   0.000    -.7093399
>    -.3769659
                                  |
                             rhfo |
                       At least 1 |    .579714   .0298135    19.44   0.000     .5212807
>     .6381473
                Not applicable    |  -.1587965   .1145888    -1.39   0.166    -.3833863
>     .0657934
                                  |
                             ghto |
                       At least 1 |   -.561311   .0688702    -8.15   0.000    -.6962941
>     -.426328
                Not applicable    |   .1504519   .0298307     5.04   0.000     .0919848
>      .208919
                                  |
                       rhfo#ghto  |
          At least 1#At least 1   |  -.0194475   .1466053    -0.13   0.894    -.3067885
>     .2678935
      At least 1#Not applicable   |   .0749786   .0318121     2.36   0.018     .0126281
>     .1373291
                                  |
                          country |
                          Austria |  -.1139314   .2090287    -0.55   0.586      -.52362
>     .2957573
                          Germany |   .0597746   .1626642     0.37   0.713    -.2590413
```

```
>     .3785906
                  Netherlands |  -.4235259   .1563104    -2.71   0.007    -.7298886
>    -.1171632
                        Spain |   .4855285   .1590372     3.05   0.002     .1738214
>     .7972356
                        Italy |  -.0617605   .1428227    -0.43   0.665    -.3416878
>     .2181668
                       France |   .2484527   .1641769     1.51   0.130     -.073328
>     .5702335
                      Denmark |   .0110323   .1973595     0.06   0.955    -.3757851
>     .3978498
                  Switzerland |  -.4761999   .1872876    -2.54   0.011    -.8432768
>     -.109123
                      Belgium |   -.486136   .0873406    -5.57   0.000    -.6573204
>    -.3149516
              Czech Republic |  -.0177938   .1856593    -0.10   0.924    -.3816794
>     .3460918
                     Slovenia |   .3516922    .125626     2.80   0.005     .1054699
>     .5979145
                              |
                        _cons |  -8.435562   1.776992    -4.75   0.000     -11.9184
>    -4.952721
------------------------------+----------------------------------------------------
  ____________

WARNING! 7 in-sample cases have an outcome with a predicted probability that is
less than 0. See the gologit2 help section on Warning Messages for more information.
results_imp1_go_3.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.748e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.703e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.239e+08  (not concave)
Iteration 3:  Log pseudolikelihood = -1.047e+08  (not concave)
Iteration 4:  Log pseudolikelihood =  -88870429  (not concave)
Iteration 5:  Log pseudolikelihood =  -85002753  (not concave)
Iteration 6:  Log pseudolikelihood =  -83121026  (not concave)
Iteration 7:  Log pseudolikelihood =  -81855984  (not concave)
Iteration 8:  Log pseudolikelihood =  -81030287  (not concave)
Iteration 9:  Log pseudolikelihood =  -80607528  (not concave)
Iteration 10: Log pseudolikelihood =  -80019943  (not concave)
Iteration 11: Log pseudolikelihood =  -79607366  (not concave)
Iteration 12: Log pseudolikelihood =  -79450085  (not concave)
Iteration 13: Log pseudolikelihood =  -79305056  (not concave)
Iteration 14: Log pseudolikelihood =  -79069435  (not concave)
Iteration 15: Log pseudolikelihood =  -78866137
Iteration 16: Log pseudolikelihood =  -78524498  (not concave)
Iteration 17: Log pseudolikelihood =  -78223184
Iteration 18: Log pseudolikelihood =  -77962586  (not concave)
Iteration 19: Log pseudolikelihood =  -77897247
Iteration 20: Log pseudolikelihood =  -77836204
Iteration 21: Log pseudolikelihood =  -77822895
Iteration 22: Log pseudolikelihood =  -77814330
Iteration 23: Log pseudolikelihood =  -77813986
Iteration 24: Log pseudolikelihood =  -77813985

Stereotype logistic regression                      Number of obs =  29,989
                                                    Wald chi2(35) = 2191.81
Log pseudolikelihood = -77813985                    Prob > chi2   =  0.0000
```

(1) **[phi1_1]_cons = 1**

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.2855644	.1310582	2.18	0.029	.0286951	.5424338
c.age#c.age	-.0014896	.0008982	-1.66	0.097	-.00325	.0002707
yedu	-.0189961	.0387601	-0.49	0.624	-.0949646	.0569723
c.yedu#c.yedu	-.0048633	.0018044	-2.70	0.007	-.0083998	-.0013267
thinc	-.4308178	.1134721	-3.80	0.000	-.653219	-.2084165
c.thinc#c.thinc	.034657	.0522985	0.66	0.508	-.0678462	.1371602
gender						
Female	.0567861	.1077934	0.53	0.598	-.154485	.2680573
0.covid	0	(omitted)				
household						
Couple, both responding	.3268993	.1433339	2.28	0.023	.04597	.6078285
Multiple, at least 1 responding	.6671401	.192907	3.46	0.001	.2890492	1.045231
cjs						
Not applicable	-.0126432	.8602519	-0.01	0.988	-1.698706	1.67342
Retired	.6287888	.2101113	2.99	0.003	.2169782	1.040599
Unemployed	-.1695894	.4134722	-0.41	0.682	-.97998	.6408012
Permanently sick or disabled	3.040167	.4475546	6.79	0.000	2.162976	3.917358
Homemaker	.4395525	.2435504	1.80	0.071	-.0377974	.9169025
Other	-1.684546	.5038557	-3.34	0.001	-2.672085	-.6970075
gali						
Limited	5.355019	.1562574	34.27	0.000	5.048761	5.661278
lifesat	-.5393119	.1860989	-2.90	0.004	-.9040591	-.1745647
c.lifesat#c.lifesat	-.0166145	.0130868	-1.27	0.204	-.0422642	.0090352
naly						
At least 1	-.9379329	.1404119	-6.68	0.000	-1.213135	-.6627306
Not applicable	0	(omitted)				
rhfo						
At least 1	1.212093	.1775195	6.83	0.000	.864161	1.560025
ghto						
At least 1	-.486821	.1546552	-3.15	0.002	-.7899396	-.1837023

```
                 Not applicable |  -.0409814   .1240095    -0.33   0.741    -.2840356
>     .2020728                  |
                                |
                      rhfo#ghto |
          At least 1#At least 1 |  -1.052221   .3349872    -3.14   0.002    -1.708784
>    -.3956584
      At least 1#Not applicable |  -.0207284   .2888545    -0.07   0.943    -.5868728
>      .545416                  |
                                |
                        country |
                        Austria |  -.2683262   .1932441    -1.39   0.165    -.6470777
>     .1104254
                        Germany |   1.095569    .221315     4.95   0.000     .6617999
>     1.529339
                    Netherlands |  -.1538249   .2075345    -0.74   0.459    -.5605851
>     .2529352
                          Spain |    1.99447   .2232099     8.94   0.000     1.556987
>     2.431953
                          Italy |   1.063176   .2137527     4.97   0.000     .6442279
>     1.482123
                         France |   .9276905    .192867     4.81   0.000     .5496781
>     1.305703
                        Denmark |  -1.315511   .2403764    -5.47   0.000     -1.78664
>    -.8443817
                    Switzerland |  -.5950807   .2187884    -2.72   0.007    -1.023898
>    -.1662633
                        Belgium |  -.0030569   .1936298    -0.02   0.987    -.3825643
>     .3764506
                 Czech Republic |   .8181709   .2325657     3.52   0.000     .3623506
>     1.273991
                       Slovenia |   1.448105   .2781214     5.21   0.000     .9029974
>     1.993213
-------------------------------+----------------------------------------------------------
  ____________
                         /phi1_1 |          1  (constrained)
                         /phi1_2 |   .8511869    .018039    47.19   0.000     .8158312
>     .8865426
                         /phi1_3 |   .6247508   .0135556    46.09   0.000     .5981823
>     .6513194
                         /phi1_4 |   .3450519   .0121792    28.33   0.000     .3211811
>     .3689227
                         /phi1_5 |          0  (base outcome)
-------------------------------+----------------------------------------------------------
  ____________
                         /theta1 |   9.938727   4.803843     2.07   0.039     .5233672
>     19.35409
                         /theta2 |   9.843153   4.094938     2.40   0.016     1.817221
>     17.86908
                         /theta3 |   9.040216   3.025275     2.99   0.003     3.110786
>     14.96965
                         /theta4 |   5.760746   1.693298     3.40   0.001     2.441943
>     9.079549
                         /theta5 |          0  (base outcome)
-------------------------------+----------------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_11_3.xls
dir : seeout

Predictive margins                                      Number of obs = 29,989
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]
_predict#rhfo#ghto					
1#None#None	.052812	.0021221	24.89	0.000	.0486528 .0569712
1#None#At least 1	.0612619	.0030427	20.13	0.000	.0552984 .0672255
1#None#Not applicable	.0534892	.0023972	22.31	0.000	.0487907 .0581876
1#At least 1#None	.0354734	.0025257	14.04	0.000	.0305231 .0404238
1#At least 1#At least 1	.0583885	.005024	11.62	0.000	.0485415 .0682354
1#At least 1#Not applicable	.036236	.0031587	11.47	0.000	.030045 .042427
2#None#None	.1326115	.003415	38.83	0.000	.1259182 .1393048
2#None#At least 1	.1456585	.0047459	30.69	0.000	.1363567 .1549603
2#None#Not applicable	.1336941	.0038284	34.92	0.000	.1261906 .1411977
2#At least 1#None	.1022107	.0046084	22.18	0.000	.0931785 .1112429
2#At least 1#At least 1	.1413311	.0075384	18.75	0.000	.1265561 .156106
2#At least 1#Not applicable	.1036735	.0058465	17.73	0.000	.0922145 .1151324
3#None#None	.3888111	.0048968	79.40	0.000	.3792135 .3984087
3#None#At least 1	.398455	.0054394	73.25	0.000	.3877941 .409116
3#None#Not applicable	.3896927	.005155	75.59	0.000	.379589 .3997963
3#At least 1#None	.3574477	.0061737	57.90	0.000	.3453474 .3695481
3#At least 1#At least 1	.39549	.0067688	58.43	0.000	.3822234 .4087566
3#At least 1#Not applicable	.3592733	.0077571	46.32	0.000	.3440696 .374477
4#None#None	.3093002	.0046508	66.50	0.000	.3001847 .3184156
4#None#At least 1	.2948122	.0057133	51.60	0.000	.2836142 .3060101
4#None#Not applicable	.3081271	.0050548	60.96	0.000	.2982198 .3180344
4#At least 1#None	.3394908	.006281	54.05	0.000	.3271802 .3518013
4#At least 1#At least 1	.2996991	.0088333	33.93	0.000	.2823861 .317012
4#At least 1#Not applicable	.3381822	.0068624	49.28	0.000	.3247322 .3516321
5#None#None	.1164653	.0036649	31.78	0.000	.1092821 .1236484
5#None#At least 1	.0998124	.0050419	19.80	0.000	.0899304 .1096943
5#None#Not applicable	.114997	.0043475	26.45	0.000	.1064761 .1235178
5#At least 1#None	.1653773	.0071609	23.09	0.000	.1513421 .1794125
5#At least 1#At least 1	.1050914	.0089011	11.81	0.000	.0876456 .1225372
5#At least 1#Not applicable	.162635	.0102006	15.94	0.000	.1426423 .1826278

```
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.578e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -2.163e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.384e+08  (not concave)
Iteration 3:  Log pseudolikelihood =  -86621299  (not concave)
Iteration 4:  Log pseudolikelihood =  -67984553  (not concave)
Iteration 5:  Log pseudolikelihood =  -62762060  (not concave)
Iteration 6:  Log pseudolikelihood =  -61665389  (not concave)
Iteration 7:  Log pseudolikelihood =  -60672275  (not concave)
Iteration 8:  Log pseudolikelihood =  -60085018  (not concave)
Iteration 9:  Log pseudolikelihood =  -59614254  (not concave)
Iteration 10: Log pseudolikelihood =  -59237207  (not concave)
Iteration 11: Log pseudolikelihood =  -58963471  (not concave)
Iteration 12: Log pseudolikelihood =  -58769069  (not concave)
Iteration 13: Log pseudolikelihood =  -58606353
Iteration 14: Log pseudolikelihood =  -58126514  (not concave)
Iteration 15: Log pseudolikelihood =  -57972417  (not concave)
Iteration 16: Log pseudolikelihood =  -57895908
Iteration 17: Log pseudolikelihood =  -57656430  (not concave)
Iteration 18: Log pseudolikelihood =  -57296067
Iteration 19: Log pseudolikelihood =  -57227819  (not concave)
Iteration 20: Log pseudolikelihood =  -57197398
Iteration 21: Log pseudolikelihood =  -57189951
Iteration 22: Log pseudolikelihood =  -57188683
Iteration 23: Log pseudolikelihood =  -57188670
Iteration 24: Log pseudolikelihood =  -57188670

Stereotype logistic regression                        Number of obs =  22,899
                                                      Wald chi2(35) = 1179.73
Log pseudolikelihood = -57188670                      Prob > chi2   =  0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------------------------------------------------------
             |               Robust
       sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]
-------------+----------------------------------------------------------------
         age |   .4563206   .1587741     2.87   0.004     .1451292
>     .767512
             |
   c.age#c.age |  -.0027267   .0010653    -2.56   0.010    -.0048147
>   -.0006387
             |
        yedu |  -.1065075   .0450405    -2.36   0.018    -.1947852
>   -.0182297
             |
 c.yedu#c.yedu |  -.0002453   .0020763    -0.12   0.906    -.0043148
>    .0038243
             |
       thinc |  -.7069291   .1355764    -5.21   0.000     -.972654
>   -.4412042
             |
 c.thinc#c.thinc |   .1716941   .0637263     2.69   0.007     .0467928
>    .2965953
             |
      gender |
     Female  |   .0553458   .1233904     0.45   0.654    -.1864948
>    .2971865
     0.covid |          0  (omitted)
             |
   household |
Couple, both responding |   .4494058    .154418     2.91   0.004     .1467522
>    .7520594
Multiple, at least 1 responding |    .482386   .2330964     2.07   0.039     .0255254
>    .9392466
             |
         cjs |
Not applicable |   1.647207   .8081072     2.04   0.042     .0633457
```

```
>     3.231068
                        Retired |   .8786243   .2904073     3.03   0.002     .3094364
>     1.447812
                     Unemployed |   .3215595   .6560006     0.49   0.624     -.964178
>     1.607297
   Permanently sick or disabled |   3.270099   .5604623     5.83   0.000     2.171613
>     4.368585
                      Homemaker |    1.02096    .337729     3.02   0.003     .3590233
>     1.682897
                          Other |   .1003248   .5347888     0.19   0.851     -.947842
>     1.148491
                                |
                           gali |
                        Limited |    5.05141   .1967168    25.68   0.000     4.665852
>     5.436968
                        lifesat |  -.7118602   .1806996    -3.94   0.000    -1.066025
>    -.3576955
                                |
             c.lifesat#c.lifesat |  -.0002093   .0130014    -0.02   0.987    -.0256916
>      .025273
                                |
                           naly |
                      At least 1 |  -.7316825   .1608024    -4.55   0.000    -1.046849
>    -.4165156
                  Not applicable |          0  (omitted)
                                |
                           rhfo |
                      At least 1 |   1.106748    .195655     5.66   0.000     .7232715
>     1.490225
                                |
                           ghto |
                      At least 1 |  -.7737881    .184571    -4.19   0.000    -1.135541
>    -.4120357
                  Not applicable |   -.125684   .1413977    -0.89   0.374    -.4028183
>     .1514503
                                |
                       rhfo#ghto |
          At least 1#At least 1 |  -.7329118   .4075875    -1.80   0.072    -1.531769
>     .0659451
      At least 1#Not applicable |   .0680092   .3508333     0.19   0.846    -.6196113
>     .7556298
                                |
                        country |
                        Austria |   .7488194   .2136901     3.50   0.000     .3299945
>     1.167644
                        Germany |   2.029033   .2530542     8.02   0.000     1.533056
>      2.52501
                    Netherlands |   .5260877   .2248675     2.34   0.019     .0853554
>     .9668199
                          Spain |   2.324455   .2585233     8.99   0.000     1.817759
>     2.831152
                          Italy |   1.759706   .2454295     7.17   0.000     1.278673
>     2.240739
                         France |   1.755819   .2148456     8.17   0.000      1.33473
>     2.176909
                        Denmark |  -.2141746   .2573337    -0.83   0.405    -.7185393
>     .2901902
                    Switzerland |   .2830831   .2375641     1.19   0.233     -.182534
>     .7487002
                        Belgium |   .5531693   .2070951     2.67   0.008     .1472703
>     .9590683
                  Czech Republic |   1.540651   .2728192     5.65   0.000     1.005935
>     2.075367
                        Slovenia |   1.405008   .3012273     4.66   0.000     .8146136
>     1.995403
------------------------------------------------------------------------------------------
                          /phi1_1 |          1  (constrained)
                          /phi1_2 |   .851071   .0265209    32.09   0.000      .799091
>      .903051
                          /phi1_3 |   .627127   .0199731    31.40   0.000     .5879804
>     .6662737
```

```
                        /phi1_4 |   .3325939   .0157128    21.17   0.000     .3017975
>     .3633904
                        /phi1_5 |          0  (base outcome)
------------------------------------------------------------------------------
  ____________
                        /theta1 |   16.16052   5.866234     2.75   0.006     4.662918
>     27.65813
                        /theta2 |   15.10741   5.020372     3.01   0.003     5.267659
>     24.94716
                        /theta3 |   12.95571   3.721079     3.48   0.000     5.662529
>     20.24889
                        /theta4 |   7.641054   2.000016     3.82   0.000     3.721095
>     11.56101
                        /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_13_3.xls
dir : seeout

Predictive margins                                      Number of obs = 22,899
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------------------------------------------------
  ________
                             |            Delta-method
                             |     Margin   std. err.      z    P>|z|     [95% conf. i
> nterval]
------------------------------------------------------------------------------
  ________
          _predict#rhfo#ghto |
                 1#None#None |   .0469663   .0022435    20.93   0.000     .0425691
> .0513635
          1#None#At least 1  |   .0599279   .0036229    16.54   0.000     .0528272
> .0670286
      1#None#Not applicable  |   .0489184   .0025709    19.03   0.000     .0438795
> .0539573
          1#At least 1#None  |   .0321694   .0026697    12.05   0.000     .0269369
> .0374018
    1#At least 1#At least 1  |   .0533821   .0057299     9.32   0.000     .0421516
> .0646126
1#At least 1#Not applicable  |   .0328396   .0037127     8.85   0.000     .0255629
> .0401163
                 2#None#None |    .122093   .0038289    31.89   0.000     .1145885
> .1295975
          2#None#At least 1  |   .1426547   .0053788    26.52   0.000     .1321123
>  .153197
      2#None#Not applicable  |   .1253533   .0043895    28.56   0.000     .1167501
> .1339566
          2#At least 1#None  |   .0950121   .0050106    18.96   0.000     .0851916
> .1048326
    2#At least 1#At least 1  |   .1325803   .0091681    14.46   0.000     .1146112
> .1505494
2#At least 1#Not applicable  |    .096344   .0072415    13.30   0.000     .0821508
> .1105372
                 3#None#None |   .3920526   .0055428    70.73   0.000     .3811888
> .4029163
          3#None#At least 1  |   .4097365   .0066069    62.02   0.000     .3967872
> .4226859
      3#None#Not applicable  |   .3952453   .0059011    66.98   0.000     .3836793
> .4068113
          3#At least 1#None  |   .3591687   .0074413    48.27   0.000      .344584
> .3737533
    3#At least 1#At least 1  |   .4017897   .0090646    44.33   0.000     .3840234
> .4195561
3#At least 1#Not applicable  |   .3610757   .0105937    34.08   0.000     .3403125
```

```
>  .381839
                 4#None#None |   .3156154   .0054049    58.39   0.000     .3050221
> .3262087
           4#None#At least 1 |   .2912922   .0068067    42.80   0.000     .2779514
>  .304633
       4#None#Not applicable |    .311877   .0057447    54.29   0.000     .3006176
> .3231363
           4#At least 1#None |   .3443153   .0073437    46.89   0.000      .329922
> .3587085
     4#At least 1#At least 1 |    .303421     .01112    27.29   0.000     .2816262
> .3252159
4#At least 1#Not applicable |   .3430224   .0087605    39.16   0.000     .3258521
> .3601927
                 5#None#None |   .1232728   .0042842    28.77   0.000     .1148758
> .1316697
           5#None#At least 1 |   .0963887   .0060169    16.02   0.000     .0845959
> .1081816
       5#None#Not applicable |    .118606   .0051963    22.83   0.000     .1084215
> .1287905
           5#At least 1#None |   .1693346    .008074    20.97   0.000     .1535098
> .1851594
     5#At least 1#At least 1 |   .1088268   .0114658     9.49   0.000     .0863542
> .1312994
5#At least 1#Not applicable |   .1667183   .0132283    12.60   0.000     .1407912
> .1926453
----------------------------+----------------------------------------------------
```

slog13_3.xls
dir : seeout
note: **0.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -1.943e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.342e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -71144921  (not concave)
Iteration 3:  Log pseudolikelihood =  -53866295  (not concave)
Iteration 4:  Log pseudolikelihood =  -49085705  (not concave)
Iteration 5:  Log pseudolikelihood =  -48376120  (not concave)
Iteration 6:  Log pseudolikelihood =  -48192781
Iteration 7:  Log pseudolikelihood =  -48140230  (not concave)
Iteration 8:  Log pseudolikelihood =  -47896419  (not concave)
Iteration 9:  Log pseudolikelihood =  -47659281
Iteration 10: Log pseudolikelihood =  -47306787  (not concave)
Iteration 11: Log pseudolikelihood =  -47173290  (not concave)
Iteration 12: Log pseudolikelihood =  -47083088  (not concave)
Iteration 13: Log pseudolikelihood =  -47009155  (not concave)
Iteration 14: Log pseudolikelihood =  -46948639  (not concave)
Iteration 15: Log pseudolikelihood =  -46903059  (not concave)
Iteration 16: Log pseudolikelihood =  -46868509  (not concave)
Iteration 17: Log pseudolikelihood =  -46839627
Iteration 18: Log pseudolikelihood =  -46725734  (not concave)
Iteration 19: Log pseudolikelihood =  -46701250
Iteration 20: Log pseudolikelihood =  -46623818  (not concave)
Iteration 21: Log pseudolikelihood =  -46603555
Iteration 22: Log pseudolikelihood =  -46601704
Iteration 23: Log pseudolikelihood =  -46596878
Iteration 24: Log pseudolikelihood =  -46596681
Iteration 25: Log pseudolikelihood =  -46596679
Iteration 26: Log pseudolikelihood =  -46596679

Stereotype logistic regression                      Number of obs =  18,545
                                                    Wald chi2(32) = 1092.78
Log pseudolikelihood = -46596679                    Prob > chi2   =  0.0000
```

(1) **[phi1_1]_cons = 1**

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.6101674	.171959	3.55	0.000	.273134	.9472008
c.age#c.age	-.0035922	.0011235	-3.20	0.001	-.0057941	-.0013902
yedu	-.0383718	.0451107	-0.85	0.395	-.1267871	.0500434
c.yedu#c.yedu	-.0024321	.002047	-1.19	0.235	-.0064441	.00158
thinc	-.4542006	.1582254	-2.87	0.004	-.7643166	-.1440845
c.thinc#c.thinc	.0079136	.0712431	0.11	0.912	-.1317204	.1475476
gender						
Female	.0774256	.1388441	0.56	0.577	-.1947037	.349555
0.covid	0	(omitted)				
household						
Couple, both responding	.3501415	.1550421	2.26	0.024	.0462646	.6540184
Multiple, at least 1 responding	.5806604	.2653434	2.19	0.029	.0605968	1.100724
cjs						
Not applicable	2.94189	.8397639	3.50	0.000	1.295983	4.587797
Retired	1.51504	.4517888	3.35	0.001	.6295506	2.40053
Unemployed	.354817	.8555072	0.41	0.678	-1.321946	2.03158
Permanently sick or disabled	3.352745	.7335058	4.57	0.000	1.9151	4.79039
Homemaker	1.715084	.4906288	3.50	0.000	.7534689	2.676698
Other	1.573913	.5464114	2.88	0.004	.5029666	2.64486
gali						
Limited	5.276877	.2071256	25.48	0.000	4.870919	5.682836
lifesat	-1.066337	.1791332	-5.95	0.000	-1.417432	-.7152428
c.lifesat#c.lifesat	.0201997	.0128848	1.57	0.117	-.0050541	.0454535
naly						
At least 1	-.3267714	.1573064	-2.08	0.038	-.6350862	-.0184565
Not applicable	0	(omitted)				
rhfo						
At least 1	1.29993	.1771607	7.34	0.000	.9527014	1.647158
ghto						
At least 1	-.7225946	.1933507	-3.74	0.000	-1.101555	-.3436341

```
                         |
            rhfo#ghto    |
  At least 1#At least 1  |  -1.097396   .3542878    -3.10   0.002    -1.791787
>    -.4030045           |
                         |
             country     |
            Austria      |   .5651829   .2151945     2.63   0.009     .1434093
>     .9869564           |
            Germany      |   1.849305   .2541058     7.28   0.000     1.351266
>     2.347343           |
              Spain      |   2.269602   .2605191     8.71   0.000     1.758994
>      2.78021           |
              Italy      |   1.774977   .2438137     7.28   0.000     1.297111
>     2.252843           |
             France      |   1.647613   .2174695     7.58   0.000     1.221381
>     2.073846           |
            Denmark      |  -.4730794   .2717085    -1.74   0.082    -1.005618
>     .0594595           |
        Switzerland      |   .3159217   .2533579     1.25   0.212    -.1806506
>      .812494           |
            Belgium      |   .3087472   .2099172     1.47   0.141    -.1026829
>     .7201773           |
      Czech Republic     |    1.86024   .2762164     6.73   0.000     1.318866
>     2.401614           |
           Slovenia      |   1.942192   .3033762     6.40   0.000     1.347586
>     2.536798           |
-------------------------+----------------------------------------------------------
             /phi1_1     |          1  (constrained)
             /phi1_2     |  .8793397   .0258058    34.08   0.000     .8287613
>     .9299181           |
             /phi1_3     |  .6587222   .0196593    33.51   0.000     .6201906
>     .6972538           |
             /phi1_4     |  .3531294   .0167451    21.09   0.000     .3203097
>     .3859492           |
             /phi1_5     |          0  (base outcome)
-------------------------+----------------------------------------------------------
             /theta1     |  22.90986   6.557258     3.49   0.000     10.05787
>     35.76185           |
             /theta2     |  21.56509    5.75846     3.74   0.000     10.27872
>     32.85146           |
             /theta3     |  17.99108   4.344326     4.14   0.000     9.476358
>      26.5058           |
             /theta4     |   10.5266   2.391842     4.40   0.000     5.838676
>     15.21452           |
             /theta5     |          0  (base outcome)
-------------------------+----------------------------------------------------------
(sphus=Poor is the base outcome)
results_imp1_ste_15_3.xls
dir : seeout

Predictive margins                                      Number of obs = 18,545
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. inter val]
_predict#rhfo#ghto					
1#None#None	.0422445	.0024076	17.55	0.000	.0375258 .0469633
1#None#At least 1	.0523678	.0033403	15.68	0.000	.0458209 .0589147
1#At least 1#None	.027713	.0023823	11.63	0.000	.0230437 .0323823
1#At least 1#At least 1	.049374	.004679	10.55	0.000	.0402035 .0585446
2#None#None	.1241934	.0040686	30.52	0.000	.116219 .1321678
2#None#At least 1	.1441262	.0061131	23.58	0.000	.1321446 .1561077
2#At least 1#None	.0918553	.0049005	18.74	0.000	.0822505 .1014602
2#At least 1#At least 1	.1384158	.0080309	17.24	0.000	.1226756 .154156
3#None#None	.3879319	.0062545	62.02	0.000	.3756733 .4001905
3#None#At least 1	.4063726	.0079486	51.12	0.000	.3907935 .4219516
3#At least 1#None	.3462428	.0070268	49.27	0.000	.3324706 .3600151
3#At least 1#At least 1	.401592	.0088507	45.37	0.000	.3842448 .4189391
4#None#None	.3321702	.0058627	56.66	0.000	.3206795 .3436609
4#None#At least 1	.3081967	.0079248	38.89	0.000	.2926645 .323729
4#At least 1#None	.3663013	.007515	48.74	0.000	.3515722 .3810303
4#At least 1#At least 1	.315221	.0102821	30.66	0.000	.2950684 .3353736
5#None#None	.1134599	.0042328	26.80	0.000	.1051638 .121756
5#None#At least 1	.0889367	.0063279	14.05	0.000	.0765342 .1013392
5#At least 1#None	.1678875	.0074784	22.45	0.000	.1532301 .182545
5#At least 1#At least 1	.0953972	.0089645	10.64	0.000	.077827 .1129673

slog15_3.xls
dir : seeout
note: **0.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.
note: **2.ghto** omitted because of collinearity.
note: **0.rhfo#2.ghto** identifies no observations in the sample.
note: **1.rhfo#2.ghto** identifies no observations in the sample.
note: **2.rhfo#0.ghto** identifies no observations in the sample.
note: **2.rhfo#1.ghto** identifies no observations in the sample.
note: **2.rhfo#2.ghto** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -2.073e+08  (not concave)
Iteration 1:  Log pseudolikelihood =  -96847925  (not concave)
Iteration 2:  Log pseudolikelihood =  -55839048  (not concave)
Iteration 3:  Log pseudolikelihood =  -44904232  (not concave)
Iteration 4:  Log pseudolikelihood =  -43766917  (not concave)
Iteration 5:  Log pseudolikelihood =  -42718015  (not concave)
Iteration 6:  Log pseudolikelihood =  -42427780  (not concave)
Iteration 7:  Log pseudolikelihood =  -42155603  (not concave)
Iteration 8:  Log pseudolikelihood =  -42026903
Iteration 9:  Log pseudolikelihood =  -41929336  (not concave)
Iteration 10: Log pseudolikelihood =  -41846333  (not concave)
Iteration 11: Log pseudolikelihood =  -41819807
Iteration 12: Log pseudolikelihood =  -41798418  (not concave)
Iteration 13: Log pseudolikelihood =  -41761973
Iteration 14: Log pseudolikelihood =  -41755319
Iteration 15: Log pseudolikelihood =  -41755275
Iteration 16: Log pseudolikelihood =  -41755275

Stereotype logistic regression                          Number of obs = 16,181
                                                        Wald chi2(33) = 866.32
Log pseudolikelihood = -41755275                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
```

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.2786944	.1768983	1.58	0.115	-.0680199	.6254087
c.age#c.age	-.0014427	.0011417	-1.26	0.206	-.0036804	.0007949
yedu	-.008805	.046458	-0.19	0.850	-.0998609	.0822509
c.yedu#c.yedu	-.003889	.0021018	-1.85	0.064	-.0080084	.0002303
thinc	-.2252847	.160089	-1.41	0.159	-.5390533	.0884839
c.thinc#c.thinc	-.2000488	.0878771	-2.28	0.023	-.3722847	-.027813
gender						
Female	-.2167409	.1372856	-1.58	0.114	-.4858158	.052334
0.covid	0	(omitted)				
household						
Couple, both responding	.1219541	.1490066	0.82	0.413	-.1700936	.4140017
Multiple, at least 1 responding	.2074354	.2746649	0.76	0.450	-.330898	.7457687
cjs						
Not applicable	3.024172	.7095912	4.26	0.000	1.633399	4.414946
Retired	1.19703	.5440238	2.20	0.028	.1307634	2.263298
Unemployed	.8867635	1.736789	0.51	0.610	-2.517281	4.290808
Permanently sick or disabled	4.308714	.8470842	5.09	0.000	2.648459	5.968968
Homemaker	1.517674	.5770259	2.63	0.009	.3867244	2.648624
Other	.9402894	.727513	1.29	0.196	-.4856098	2.366189

```
                                 |
                            gali |
                         Limited |   4.778009   .2105756    22.69   0.000     4.365288
>      5.19073
                          lifesat |  -.3059745   .2090544    -1.46   0.143    -.7157137
>     .1037646
                                 |
             c.lifesat#c.lifesat |  -.0250487   .0148656    -1.69   0.092    -.0541848
>     .0040874
                                 |
                            naly |
                       At least 1 |  -.3691077   .1674682    -2.20   0.028    -.6973392
>    -.0408761
                   Not applicable |          0  (omitted)
                                 |
                            rhfo |
                       At least 1 |   .7815339   .2300175     3.40   0.001     .3307079
>      1.23236
                   Not applicable |   .2503216    .247323     1.01   0.311    -.2344226
>     .7350658
                                 |
                            ghto |
                       At least 1 |  -.6821218   .2324877    -2.93   0.003    -1.137789
>    -.2264544
                   Not applicable |          0  (omitted)
                                 |
                        rhfo#ghto |
             None#Not applicable |          0  (empty)
          At least 1#At least 1 |   .7328942   .4243109     1.73   0.084    -.0987399
>     1.564528
      At least 1#Not applicable |          0  (empty)
            Not applicable#None |          0  (empty)
      Not applicable#At least 1 |          0  (empty)
  Not applicable#Not applicable |          0  (omitted)
                                 |
                         country |
                         Austria |   .6918976   .2252227     3.07   0.002     .2504692
>     1.133326
                         Germany |   1.589242   .2623901     6.06   0.000     1.074967
>     2.103518
                           Spain |   2.131998   .2663709     8.00   0.000      1.60992
>     2.654075
                           Italy |   1.606771   .2485447     6.46   0.000     1.119632
>     2.093909
                          France |   1.085359   .2218269     4.89   0.000     .6505868
>     1.520132
                         Denmark |  -.7723863   .2678014    -2.88   0.004    -1.297267
>    -.2475051
                     Switzerland |   .0894786   .2358969     0.38   0.704    -.3728709
>      .551828
                         Belgium |   .0025742   .2096378     0.01   0.990    -.4083083
>     .4134567
                   Czech Republic |   .2448285   .2379057     1.03   0.303    -.2214582
>     .7111152
                         Slovenia |   1.594522   .2856057     5.58   0.000     1.034745
>     2.154299
-------------------------------------------------------------------------------------------
                          /phi1_1 |          1  (constrained)
                          /phi1_2 |    .919372   .0314351    29.25   0.000     .8577604
>     .9809837
                          /phi1_3 |   .6711816   .0229098    29.30   0.000     .6262791
>      .716084
                          /phi1_4 |   .3441637   .0195748    17.58   0.000     .3057977
>     .3825296
                          /phi1_5 |          0  (base outcome)
-------------------------------------------------------------------------------------------
                          /theta1 |   12.16352    6.88675     1.77   0.077    -1.334257
>     25.66131
                          /theta2 |   12.44735   6.358935     1.96   0.050     -.015932
>     24.91064
```

```
                        /theta3 |   10.88482   4.652801     2.34   0.019     1.765502
>     20.00415
                        /theta4 |   6.400214   2.430871     2.63   0.008     1.635794
>     11.16463
                        /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_17_3.xls
dir : seeout

Predictive margins                                      Number of obs = 16,181
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------------------------------------------------
  ____________
                                |            Delta-method
                                |     Margin   std. err.      z    P>|z|     [95% con
> f. interval]
------------------------------------------------------------------------------
  ____________
             _predict#rhfo#ghto |
                   1#None#None  |   .0405927     .00262    15.49   0.000     .0354576
>     .0457277
             1#None#At least 1  |   .0497483   .0039463    12.61   0.000     .0420137
>     .0574829
         1#None#Not applicable  |          .  (not estimable)
             1#At least 1#None  |   .0316023   .0028773    10.98   0.000     .0259629
>     .0372418
        1#At least 1#At least 1 |   .0310716   .0037358     8.32   0.000     .0237497
>     .0383936
    1#At least 1#Not applicable |          .  (not estimable)
         1#Not applicable#None  |          .  (not estimable)
    1#Not applicable#At least 1 |          .  (not estimable)
1#Not applicable#Not applicable |   .0375424   .0028887    13.00   0.000     .0318806
>     .0432042
                   2#None#None  |   .1198957   .0050668    23.66   0.000      .109965
>     .1298263
             2#None#At least 1  |   .1409984   .0081079    17.39   0.000     .1251072
>     .1568896
         2#None#Not applicable  |          .  (not estimable)
             2#At least 1#None  |   .0978905   .0064975    15.07   0.000     .0851557
>     .1106253
        2#At least 1#At least 1 |   .0965459   .0085038    11.35   0.000     .0798788
>      .113213
    2#At least 1#Not applicable |          .  (not estimable)
         2#Not applicable#None  |          .  (not estimable)
    2#Not applicable#At least 1 |          .  (not estimable)
2#Not applicable#Not applicable |   .1125851    .006245    18.03   0.000     .1003451
>     .1248251
                   3#None#None  |   .3829062   .0073493    52.10   0.000     .3685019
>     .3973106
             3#None#At least 1  |   .4029697    .009521    42.32   0.000     .3843088
>     .4216305
         3#None#Not applicable  |          .  (not estimable)
             3#At least 1#None  |   .3555686   .0093024    38.22   0.000     .3373362
>      .373801
        3#At least 1#At least 1 |   .3536535   .0132504    26.69   0.000     .3276831
>     .3796238
    3#At least 1#Not applicable |          .  (not estimable)
         3#Not applicable#None  |          .  (not estimable)
    3#Not applicable#At least 1 |          .  (not estimable)
3#Not applicable#Not applicable |   .3746148   .0079086    47.37   0.000     .3591143
>     .3901153
                   4#None#None  |   .3322387    .007136    46.56   0.000     .3182523
>      .346225
```

```
              4#None#At least 1 |   .3079038   .0103154    29.85   0.000     .2876861
>     .3281216
          4#None#Not applicable |          .  (not estimable)
              4#At least 1#None |   .3558814   .0086507    41.14   0.000     .3389264
>     .3728365
        4#At least 1#At least 1 |     .35724   .0105233    33.95   0.000     .3366148
>     .3778652
    4#At least 1#Not applicable |          .  (not estimable)
          4#Not applicable#None |          .  (not estimable)
    4#Not applicable#At least 1 |          .  (not estimable)
4#Not applicable#Not applicable |   .3403468   .0077627    43.84   0.000     .3251321
>     .3555615
                     5#None#None |   .1243668   .0058942    21.10   0.000     .1128144
>     .1359192
              5#None#At least 1 |   .0983797   .0085419    11.52   0.000     .0816379
>     .1151216
          5#None#Not applicable |          .  (not estimable)
              5#At least 1#None |   .1590572   .0103159    15.42   0.000     .1388384
>     .1792759
        5#At least 1#At least 1 |   .1614891   .0154364    10.46   0.000     .1312342
>     .1917439
    5#At least 1#Not applicable |          .  (not estimable)
          5#Not applicable#None |          .  (not estimable)
    5#Not applicable#At least 1 |          .  (not estimable)
5#Not applicable#Not applicable |   .1349109   .0082548    16.34   0.000     .1187319
>      .15109
------------------------------------------------------------------------------------------
```

slog17_3.xls
dir : seeout
note: **0.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -2.278e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.553e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -83919293  (not concave)
Iteration 3:  Log pseudolikelihood =  -47784470  (not concave)
Iteration 4:  Log pseudolikelihood =  -36180935  (not concave)
Iteration 5:  Log pseudolikelihood =  -32649381  (not concave)
Iteration 6:  Log pseudolikelihood =  -31093710  (not concave)
Iteration 7:  Log pseudolikelihood =  -30595032  (not concave)
Iteration 8:  Log pseudolikelihood =  -30252978  (not concave)
Iteration 9:  Log pseudolikelihood =  -30083705  (not concave)
Iteration 10: Log pseudolikelihood =  -29886655  (not concave)
Iteration 11: Log pseudolikelihood =  -29677594  (not concave)
Iteration 12: Log pseudolikelihood =  -29585238  (not concave)
Iteration 13: Log pseudolikelihood =  -29450412  (not concave)
Iteration 14: Log pseudolikelihood =  -29325254  (not concave)
Iteration 15: Log pseudolikelihood =  -28979820  (not concave)
Iteration 16: Log pseudolikelihood =  -28912782  (not concave)
Iteration 17: Log pseudolikelihood =  -28847967  (not concave)
Iteration 18: Log pseudolikelihood =  -28788693  (not concave)
Iteration 19: Log pseudolikelihood =  -28737304  (not concave)
Iteration 20: Log pseudolikelihood =  -28710075  (not concave)
Iteration 21: Log pseudolikelihood =  -28616809  (not concave)
Iteration 22: Log pseudolikelihood =  -28572807  (not concave)
Iteration 23: Log pseudolikelihood =  -28529030  (not concave)
Iteration 24: Log pseudolikelihood =  -28508014  (not concave)
Iteration 25: Log pseudolikelihood =  -28492928  (not concave)
Iteration 26: Log pseudolikelihood =  -28478503  (not concave)
Iteration 27: Log pseudolikelihood =  -28451450  (not concave)
Iteration 28: Log pseudolikelihood =  -28433904  (not concave)
Iteration 29: Log pseudolikelihood =  -28425078  (not concave)
Iteration 30: Log pseudolikelihood =  -28409710  (not concave)
Iteration 31: Log pseudolikelihood =  -28396847  (not concave)
Iteration 32: Log pseudolikelihood =  -28380882  (not concave)
Iteration 33: Log pseudolikelihood =  -28358206  (not concave)
Iteration 34: Log pseudolikelihood =  -28349717  (not concave)
Iteration 35: Log pseudolikelihood =  -28341313  (not concave)
Iteration 36: Log pseudolikelihood =  -28270518  (not concave)
Iteration 37: Log pseudolikelihood =  -28237420  (not concave)
Iteration 38: Log pseudolikelihood =  -28229486  (not concave)
```

```
Iteration 39: Log pseudolikelihood =  -28223137  (not concave)
Iteration 40: Log pseudolikelihood =  -28217145  (not concave)
Iteration 41: Log pseudolikelihood =  -28211396  (not concave)
Iteration 42: Log pseudolikelihood =  -28205559
Iteration 43: Log pseudolikelihood =  -28070219  (not concave)
Iteration 44: Log pseudolikelihood =  -28043382
Iteration 45: Log pseudolikelihood =  -27943559  (not concave)
Iteration 46: Log pseudolikelihood =  -27931578
Iteration 47: Log pseudolikelihood =  -27912927  (not concave)
Iteration 48: Log pseudolikelihood =  -27893617
Iteration 49: Log pseudolikelihood =  -27892301
Iteration 50: Log pseudolikelihood =  -27884299
Iteration 51: Log pseudolikelihood =  -27880875
Iteration 52: Log pseudolikelihood =  -27879777
Iteration 53: Log pseudolikelihood =  -27879735
Iteration 54: Log pseudolikelihood =  -27879734

Stereotype logistic regression                          Number of obs = 10,710
                                                        Wald chi2(33) = 750.03
Log pseudolikelihood = -27879734                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
```

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]
age	.5561894	.2559164	2.17	0.030	.0546024 1.057776
c.age#c.age	-.0032657	.0016021	-2.04	0.042	-.0064057 -.0001257
yedu	-.0209754	.0610673	-0.34	0.731	-.1406651 .0987142
c.yedu#c.yedu	-.0026548	.0027449	-0.97	0.333	-.0080346 .002725
thinc	-.1836221	.1853189	-0.99	0.322	-.5468404 .1795962
c.thinc#c.thinc	-.0239351	.0890784	-0.27	0.788	-.1985256 .1506553
gender					
Female	-.0376015	.1793917	-0.21	0.834	-.3892027 .3139998
0.covid	0	(omitted)			
household					
Couple, both responding	.4987512	.2046567	2.44	0.015	.0976315 .8998709
Multiple, at least 1 responding	.493078	.3447332	1.43	0.153	-.1825868 1.168743
cjs					
Not applicable	3.830454	1.27134	3.01	0.003	1.338673 6.322236
Retired	3.845149	1.092613	3.52	0.000	1.703667 5.98663
Unemployed	2.965671	2.547732	1.16	0.244	-2.027791 7.959134
Permanently sick or disabled	5.382167	1.353148	3.98	0.000	2.730045 8.034289
Homemaker	3.414221	1.122572	3.04	0.002	1.214019 5.614422
Other	3.656289	1.202291	3.04	0.002	1.299841 6.012736

gali					
Limited	5.335044	.2927624	18.22	0.000	4.761241 > 5.908848
lifesat	-1.477118	.3183851	-4.64	0.000	-2.101142 > -.853095
c.lifesat#c.lifesat	.047212	.0221662	2.13	0.033	.0037669 > .090657
naly					
At least 1	-1.005107	.2620077	-3.84	0.000	-1.518632 > -.4915809
Not applicable	0	(omitted)			
rhfo					
At least 1	.8009427	.2267113	3.53	0.000	.3565967 > 1.245289
ghto					
At least 1	-.6073683	.2410983	-2.52	0.012	-1.079912 > -.1348243
rhfo#ghto					
At least 1#At least 1	-.2156231	.4660458	-0.46	0.644	-1.129056 > .6978099
country					
Austria	.4954742	.3025322	1.64	0.101	-.097478 > 1.088426
Germany	1.115106	.3396823	3.28	0.001	.4493406 > 1.780871
Netherlands	-.1473874	.3228165	-0.46	0.648	-.7800962 > .4853214
Spain	2.184433	.3617424	6.04	0.000	1.475431 > 2.893435
Italy	1.356963	.3583618	3.79	0.000	.6545866 > 2.059339
France	.957168	.2876017	3.33	0.001	.393479 > 1.520857
Denmark	-.4267937	.3489154	-1.22	0.221	-1.110655 > .2570679
Switzerland	-.1505228	.3358596	-0.45	0.654	-.8087956 > .50775
Belgium	.0664842	.2982328	0.22	0.824	-.5180413 > .6510097
Czech Republic	-.2709092	.3395753	-0.80	0.425	-.9364644 > .3946461
Slovenia	1.160343	.376551	3.08	0.002	.422317 > 1.89837
/phi1_1	1	(constrained)			
/phi1_2	.8497544	.0377324	22.52	0.000	.7758003 > .9237086
/phi1_3	.65176	.0281944	23.12	0.000	.5964999 > .7070201
/phi1_4	.3420968	.0230567	14.84	0.000	.2969064 > .3872872
/phi1_5	0	(base outcome)			
/theta1	20.59984	10.19363	2.02	0.043	.6207017 > 40.57899
/theta2	19.15555	8.795883	2.18	0.029	1.915933 > 36.39516
/theta3	16.41977	6.712474	2.45	0.014	3.263559 > 29.57597
/theta4	9.549247	3.600495	2.65	0.008	2.492406 > 16.60609
/theta5	0	(base outcome)			

(sphus=Poor is the base outcome)
results_imp1_ste_20_3.xls
dir : seeout

Predictive margins Number of obs = **10,710**
Model VCE: **Robust**

1._predict: **Pr(sphus==Excellent), predict(pr outcome(#1))**
2._predict: **Pr(sphus==Very good), predict(pr outcome(#2))**
3._predict: **Pr(sphus==Good), predict(pr outcome(#3))**
4._predict: **Pr(sphus==Fair), predict(pr outcome(#4))**
5._predict: **Pr(sphus==Poor), predict(pr outcome(#5))**

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]	
_predict#rhfo#ghto						
1#None#None	.0365802	.0026959	13.57	0.000	.0312963	.0418641
1#None#At least 1	.0442465	.0038507	11.49	0.000	.0366992	.0517937
1#At least 1#None	.0280576	.002808	9.99	0.000	.0225541	.0335611
1#At least 1#At least 1	.0368397	.0049603	7.43	0.000	.0271177	.0465616
2#None#None	.1292181	.0058374	22.14	0.000	.1177771	.1406592
2#None#At least 1	.1459017	.0079001	18.47	0.000	.1304177	.1613856
2#At least 1#None	.1086059	.0064556	16.82	0.000	.0959532	.1212586
2#At least 1#At least 1	.1298089	.0109067	11.90	0.000	.1084321	.1511857
3#None#None	.3872198	.0081472	47.53	0.000	.3712515	.4031881
3#None#At least 1	.4047168	.0104724	38.65	0.000	.3841914	.4252423
3#At least 1#None	.3605762	.0095994	37.56	0.000	.3417616	.3793907
3#At least 1#At least 1	.3878991	.0138155	28.08	0.000	.3608212	.4149777
4#None#None	.3424629	.0078437	43.66	0.000	.3270895	.3578363
4#None#At least 1	.3201516	.010727	29.85	0.000	.2991271	.3411176
4#At least 1#None	.3679384	.009827	37.44	0.000	.3486778	.3877199
4#At least 1#At least 1	.341694	.0143833	23.76	0.000	.3135032	.3698847
5#None#None	.104519	.0054133	19.31	0.000	.0939091	.1151288
5#None#At least 1	.0849835	.0074612	11.39	0.000	.0703598	.0996071
5#At least 1#None	.1348219	.0084969	15.87	0.000	.1181683	.1514756
5#At least 1#At least 1	.1037584	.0135627	7.65	0.000	.0771759	.1303408

slog20_3.xls
dir : seeout
note: **1.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -1.977e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.661e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.040e+08  (not concave)
Iteration 3:  Log pseudolikelihood =  -87738617  (not concave)
Iteration 4:  Log pseudolikelihood =  -61006566  (not concave)
Iteration 5:  Log pseudolikelihood =  -41828789  (not concave)
Iteration 6:  Log pseudolikelihood =  -37884185  (not concave)
Iteration 7:  Log pseudolikelihood =  -34982183  (not concave)
Iteration 8:  Log pseudolikelihood =  -33031106  (not concave)
Iteration 9:  Log pseudolikelihood =  -32335272  (not concave)
Iteration 10: Log pseudolikelihood =  -30997880  (not concave)
Iteration 11: Log pseudolikelihood =  -30387884  (not concave)
Iteration 12: Log pseudolikelihood =  -29886320  (not concave)
Iteration 13: Log pseudolikelihood =  -29660869  (not concave)
Iteration 14: Log pseudolikelihood =  -29523788  (not concave)
Iteration 15: Log pseudolikelihood =  -29394084  (not concave)
Iteration 16: Log pseudolikelihood =  -29008376  (not concave)
Iteration 17: Log pseudolikelihood =  -28802001  (not concave)
Iteration 18: Log pseudolikelihood =  -28648581  (not concave)
Iteration 19: Log pseudolikelihood =  -28584941  (not concave)
Iteration 20: Log pseudolikelihood =  -28535680  (not concave)
Iteration 21: Log pseudolikelihood =  -28493122  (not concave)
Iteration 22: Log pseudolikelihood =  -28455980  (not concave)
Iteration 23: Log pseudolikelihood =  -28422987  (not concave)
Iteration 24: Log pseudolikelihood =  -28393320
Iteration 25: Log pseudolikelihood =  -28363564  (not concave)
Iteration 26: Log pseudolikelihood =  -28321367  (not concave)
Iteration 27: Log pseudolikelihood =  -28287045
Iteration 28: Log pseudolikelihood =  -28248498  (not concave)
Iteration 29: Log pseudolikelihood =  -28193999
Iteration 30: Log pseudolikelihood =  -28188442  (not concave)
Iteration 31: Log pseudolikelihood =  -28149690
Iteration 32: Log pseudolikelihood =  -28140431
Iteration 33: Log pseudolikelihood =  -28136386
Iteration 34: Log pseudolikelihood =  -28135650
Iteration 35: Log pseudolikelihood =  -28135631
Iteration 36: Log pseudolikelihood =  -28135631

Stereotype logistic regression                          Number of obs = 11,204
                                                        Wald chi2(32) = 583.45
Log pseudolikelihood = -28135631                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
```

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.8715471	.3019549	2.89	0.004	.2797263	1.463368
c.age#c.age	-.0050417	.0018636	-2.71	0.007	-.0086943	-.0013892
yedu	.0432806	.0671955	0.64	0.520	-.08842	.1749813
c.yedu#c.yedu	-.0045105	.0029307	-1.54	0.124	-.0102545	.0012335
thinc	-.3269316	.1965379	-1.66	0.096	-.7121387	.0582756
c.thinc#c.thinc	-.0112259	.0935421	-0.12	0.904	-.194565	.1721133
gender						
Female	.1042891	.1804707	0.58	0.563	-.2494271	.4580052
1.covid	0	(omitted)				

```
                                 |
                       household |
        Couple, both responding  |   .7147783    .206483     3.46   0.001     .3100791
>     1.119478
Multiple, at least 1 responding  |   .4071114   .4192501     0.97   0.332    -.4146036
>     1.228826
                                 |
                             cjs |
                Not applicable   |  -.4208404   1.961053    -0.21   0.830    -4.264434
>     3.422753
                        Retired  |  -.3011212   1.824483    -0.17   0.869    -3.877042
>     3.274799
   Permanently sick or disabled  |   1.178849   2.085819     0.57   0.572    -2.909281
>     5.266979
                      Homemaker  |  -.7303334    1.84628    -0.40   0.692    -4.348975
>     2.888308
                          Other  |   .1578925   1.931122     0.08   0.935    -3.627037
>     3.942822
                                 |
                            gali |
                        Limited  |   5.686986   .3073331    18.50   0.000     5.084624
>     6.289348
                         lifesat |  -.9075666   .3368231    -2.69   0.007    -1.567728
>    -.2474054
                                 |
             c.lifesat#c.lifesat |  -.0016075   .0233905    -0.07   0.945    -.0474521
>     .0442371
                                 |
                            naly |
                      At least 1 |  -.9119032   .2267983    -4.02   0.000     -1.35642
>    -.4673868
                Not applicable   |          0  (omitted)
                                 |
                            rhfo |
                      At least 1 |   .8754853   .2211234     3.96   0.000     .4420914
>     1.308879
                                 |
                            ghto |
                      At least 1 |  -.5783876   .2793962    -2.07   0.038    -1.125994
>    -.0307811
                                 |
                        rhfo#ghto |
          At least 1#At least 1  |   .0486854   .5016732     0.10   0.923     -.934576
>     1.031947
                                 |
                         country |
                         Austria |   1.029049   .3073122     3.35   0.001     .4267281
>      1.63137
                         Germany |   1.696416   .3645035     4.65   0.000      .982002
>      2.41083
                     Netherlands |  -.0205114   .3427359    -0.06   0.952    -.6922613
>     .6512386
                           Spain |   2.361681   .3777021     6.25   0.000     1.621399
>     3.101964
                           Italy |   1.798219   .3549624     5.07   0.000     1.102506
>     2.493933
                          France |   1.173662   .3068925     3.82   0.000      .572164
>     1.775161
                         Denmark |  -.0555446   .3663802    -0.15   0.879    -.7736366
>     .6625473
                     Switzerland |     .32719    .373267     0.88   0.381    -.4043999
>      1.05878
                         Belgium |   .3439233   .2998013     1.15   0.251    -.2436763
>      .931523
                 Czech Republic  |   .1077797   .3771439     0.29   0.775    -.6314088
>     .8469682
                        Slovenia |   1.578451   .3596373     4.39   0.000     .8735752
>     2.283327
-------------------------------------------------------------------------------------------
                                 |
                         /phi1_1 |          1  (constrained)
                         /phi1_2 |   .8384667   .0375276    22.34   0.000     .7649139
```

```
>     .9120195
                        /phi1_3 |   .6350174   .0297317    21.36   0.000     .5767443
>     .6932904
                        /phi1_4 |   .3551743   .0242919    14.62   0.000     .3075631
>     .4027855
                        /phi1_5 |          0  (base outcome)
------------------------------------------------------------------------------
  ------------
                        /theta1 |  32.90215   12.34535     2.67   0.008     8.705696
>      57.0986
                        /theta2 |  29.17763   10.27126     2.84   0.005     9.046332
>     49.30892
                        /theta3 |  23.96352   7.790012     3.08   0.002     8.695376
>     39.23166
                        /theta4 |  14.24033   4.340445     3.28   0.001     5.733212
>     22.74744
                        /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
  ------------
(sphus=Poor is the base outcome)
results_imp1_ste_22_3.xls
dir : seeout

Predictive margins                                      Number of obs = 11,204
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------------------------------------------------
  ----
                         |            Delta-method
                         |     Margin   std. err.      z    P>|z|     [95% conf. inter
> val]
------------------------------------------------------------------------------
  ----
     _predict#rhfo#ghto  |
            1#None#None  |    .037096   .0026945    13.77   0.000     .0318149    .042
> 3772
      1#None#At least 1  |   .0446611   .0046671     9.57   0.000     .0355137    .053
> 8085
      1#At least 1#None  |   .0275599   .0027946     9.86   0.000     .0220826    .033
> 0373
1#At least 1#At least 1  |   .0330665   .0049472     6.68   0.000     .0233701    .042
> 7628
            2#None#None  |   .1158115   .0049305    23.49   0.000     .1061479    .125
> 4751
      2#None#At least 1  |   .1298115   .0077664    16.71   0.000     .1145897    .145
> 0334
      2#At least 1#None  |   .0960089   .0055786    17.21   0.000     .0850751    .106
> 9427
2#At least 1#At least 1  |   .1077768   .0094786    11.37   0.000      .089199    .126
> 3545
            3#None#None  |   .3921804   .0079718    49.20   0.000     .3765559    .407
> 8049
      3#None#At least 1  |   .4082903   .0107577    37.95   0.000     .3872056    .429
> 3749
      3#At least 1#None  |   .3640331   .0091061    39.98   0.000     .3461855    .381
> 8807
3#At least 1#At least 1  |   .3815562   .0139442    27.36   0.000     .3542261    .408
> 8862
            4#None#None  |   .3406506   .0074105    45.97   0.000     .3261263    .355
> 1749
      4#None#At least 1  |   .3231222   .0107399    30.09   0.000     .3020724    .344
> 1721
      4#At least 1#None  |   .3624221   .0089635    40.43   0.000     .3448539    .379
> 9902
4#At least 1#At least 1  |   .3499839   .0122079    28.67   0.000     .3260568    .373
> 9111
```

```
            5#None#None |   .1142615   .0053946    21.18   0.000     .1036884    .124
> 8347
      5#None#At least 1 |   .0941149   .0093284    10.09   0.000     .0758317    .112
> 3981
      5#At least 1#None |    .149976   .0086439    17.35   0.000     .1330342    .166
> 9178
5#At least 1#At least 1 |   .1276166    .015592     8.18   0.000     .0970568    .158
> 1764
                        |
------------------------+----------------------------------------------------------------
  ____
slog22_3.xls
dir : seeout
      name:  The_value_of_a_helping_hand_sup
       log:  C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_3.smcl
  log type:  smcl
 closed on:   5 Apr 2026, 14:06:39
--------------------------------------------------------------------------------------
```